

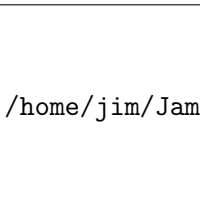
Section 4.4: The Normal Distribution

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Wednesday 5th November, 2014



/home/jim/JamesAshe/MHU/images/MarsHillU

Discrete vs Continuous Variables

Discrete vs Continuous Variables

Discrete vs Continuous Variables

Discrete

Discrete vs Continuous Variables

Discrete

ex. Number of students in class

Discrete vs Continuous Variables

Discrete

ex. Number of students in class

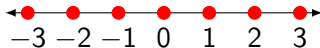
ex. Integers:

Discrete vs Continuous Variables

Discrete

ex. Number of students in class

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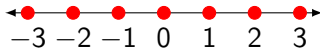


Discrete vs Continuous Variables

Discrete

ex. Number of students in class

ex. Integers:



ex. Votes for a candidate

Discrete vs Continuous Variables

Discrete

ex. Number of students in class

ex. Integers:



ex. Votes for a candidate

ex. Money: 1, .25, .10, .05, .01

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Continuous

ex. The real numbers

Discrete vs Continuous Variables

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Continuous

ex. The real numbers

ex. $-1/2 < x \leq 3$

Discrete vs Continuous Variables

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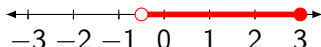
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Discrete vs Continuous Variables

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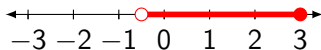
ex. Votes for a candidate

ex. Money: 1, .25, .10, .05, .01

Continuous

ex. The real numbers

ex. $-1/2 < x \leq 3$



ex. height, weight, time, etc.

Heights of Men

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The heights of
10,000 randomly
selected men.

- $n = 10,000$
- $\bar{x} = 70$ inches

The heights of
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- $\bar{x} = 70$ inches
- $s = 3$ inches

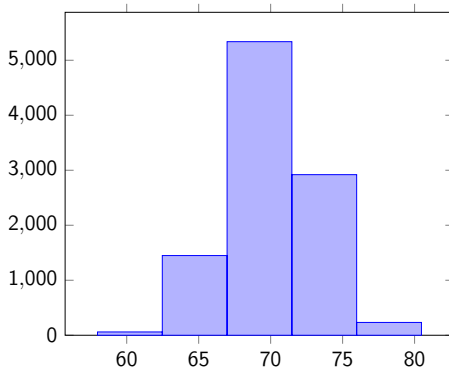
Histogram

Heights of Men

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- $s = 3$ inches

Histogram



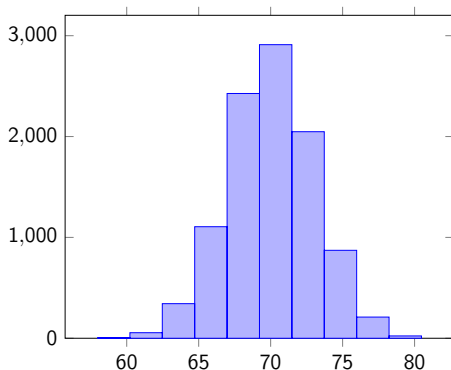
- 5 groups

Heights of Men

The heights of
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- $s = 3$ inches

Histogram



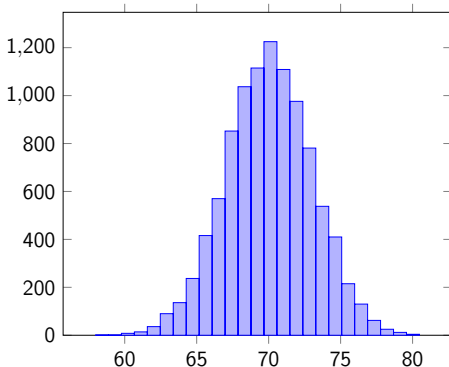
- 10 groups

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Histogram



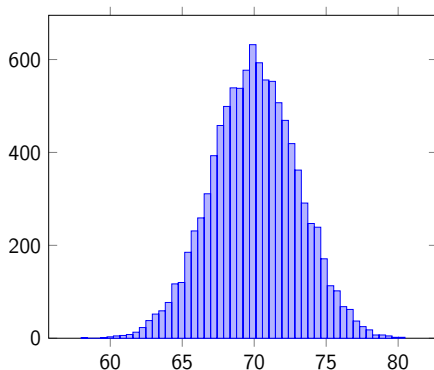
- 25 groups

Heights of Men

The heights of
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selected men.

- $n = 10,000$
- $\bar{x} = 70$ inches
- $s = 3$ inches

Histogram



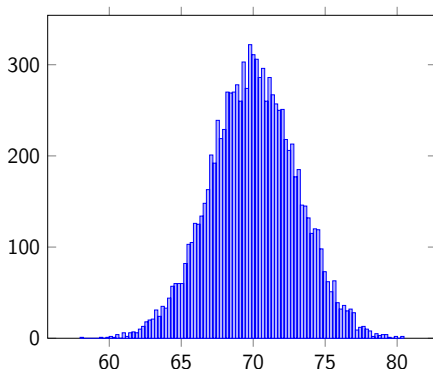
- 50 groups

Heights of Men

The heights of
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selected men.

- $n = 10,000$
- $\bar{x} = 70$ inches
- $s = 3$ inches

Histogram



- 100 groups

Standard Normal Distribution

Standard Normal Distribution

Standard Normal Distribution

Sums of 10,000
randomly generated
numbers.

- $n = 10,000$
- $\bar{x} = 0$

Standard Normal Distribution

Sums of 10,000
randomly generated
numbers.

Histogram

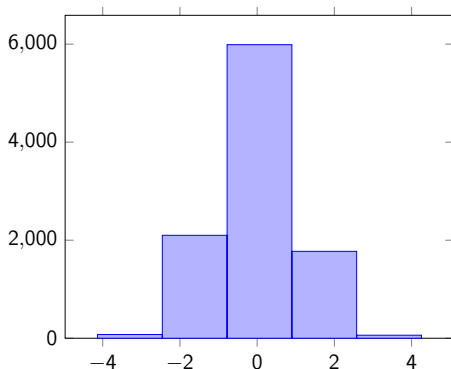
- $n = 10,000$
- $\bar{x} = 0$
- $s = 1$

Standard Normal Distribution

Sums of 10,000
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numbers.

- $n = 10,000$
- $\bar{x} = 0$
- $s = 1$

Histogram



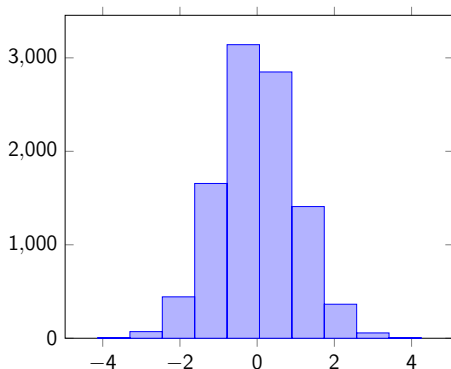
- 5 groups

Standard Normal Distribution

Sums of 10,000
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- $s = 1$

Histogram



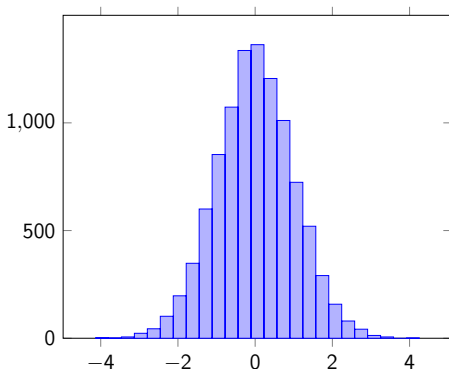
- 10 groups

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Sums of 10,000
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Histogram



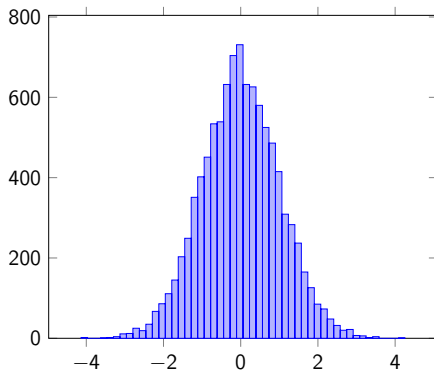
- 25 groups

Standard Normal Distribution

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Histogram



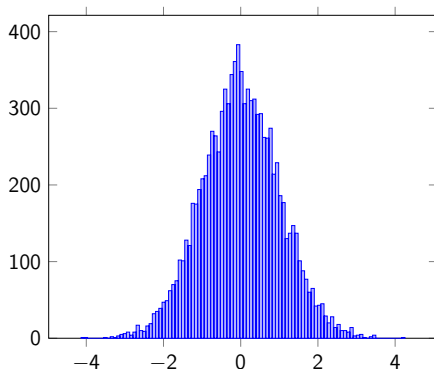
- 50 groups

Standard Normal Distribution

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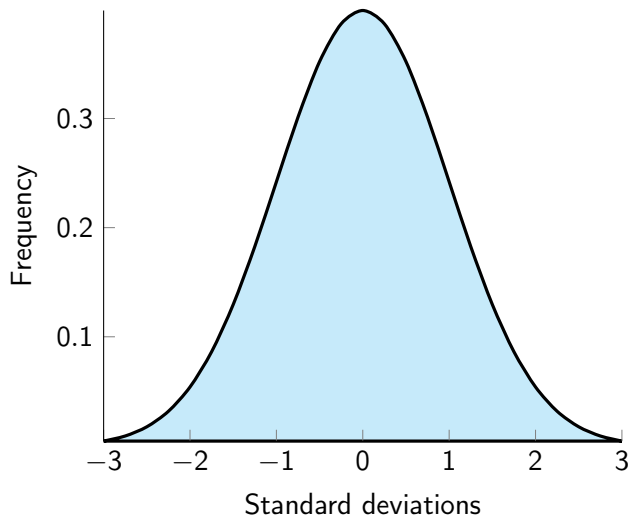
Histogram



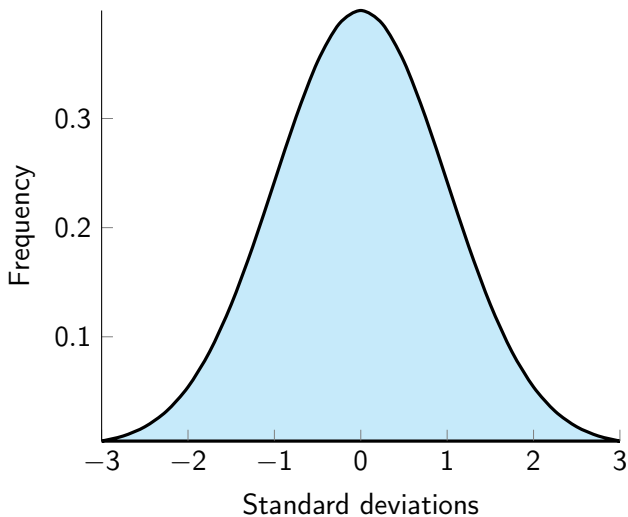
- 100 groups

Normal Distribution

Normal Distribution



Normal Distribution

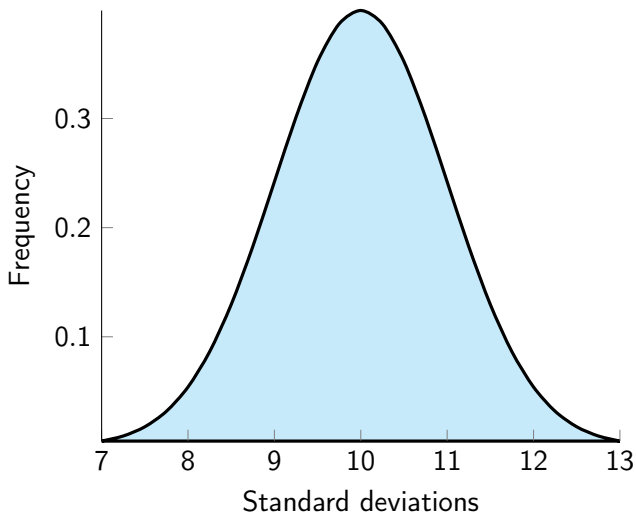


$$\bar{x} = 0$$

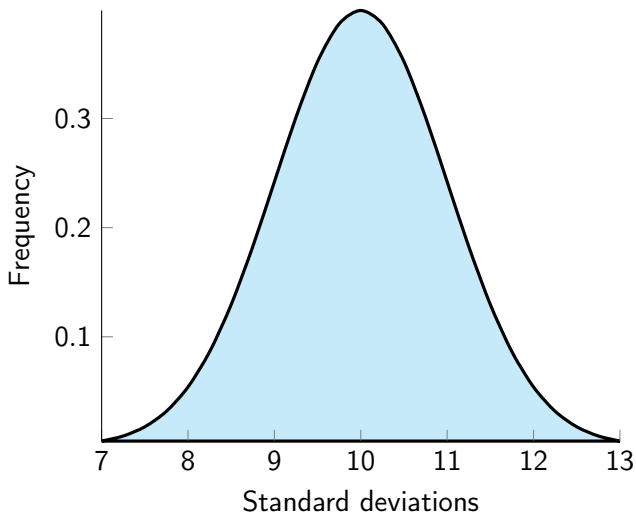
$$s = 1$$

Normal Distribution

Normal Distribution



Normal Distribution

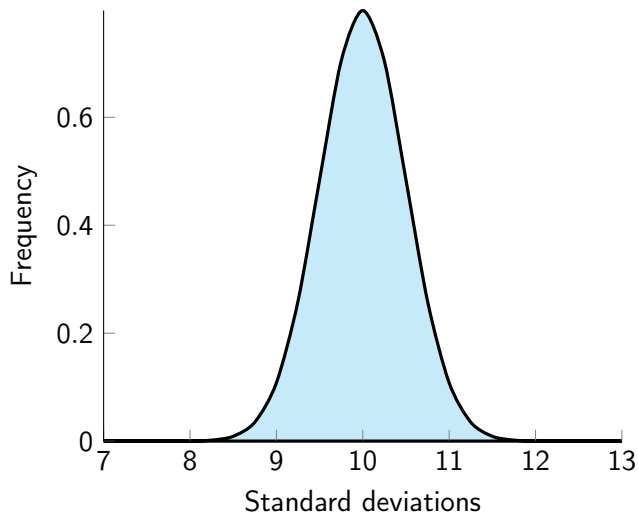


$$\bar{x} = 10$$

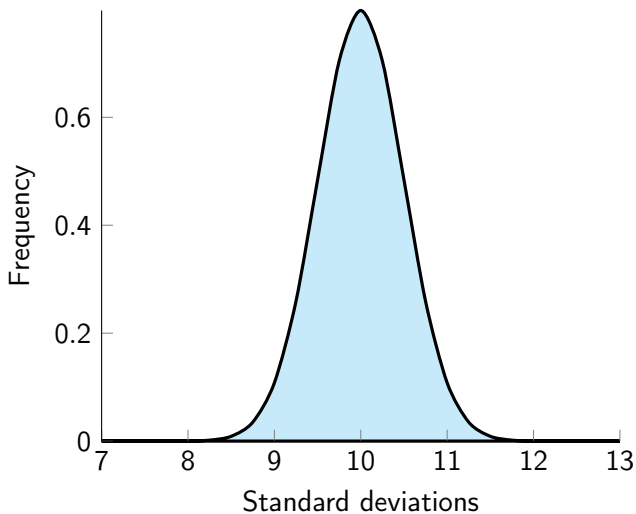
$$s = 1$$

Normal Distribution

Normal Distribution



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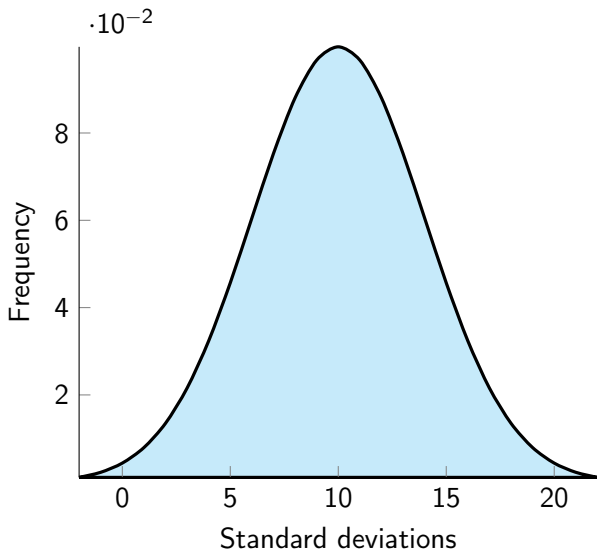


$$\bar{x} = 10$$

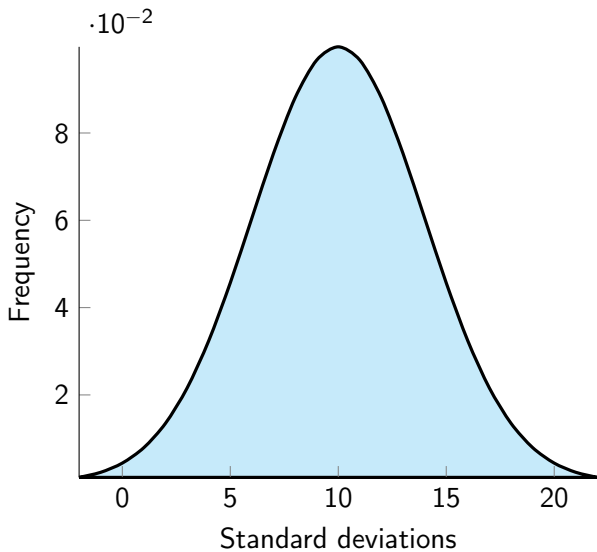
$$s = .5$$

Normal Distribution

Normal Distribution



Normal Distribution



$$\bar{x} = 10$$

$$s = 4$$

$\bar{x}$: means

Sample

$\bar{x}$: means

s : standard deviation

Sample

$\bar{x}$: means

s : standard deviation

μ : means

Sample

$\bar{x}$: means

s : standard deviation

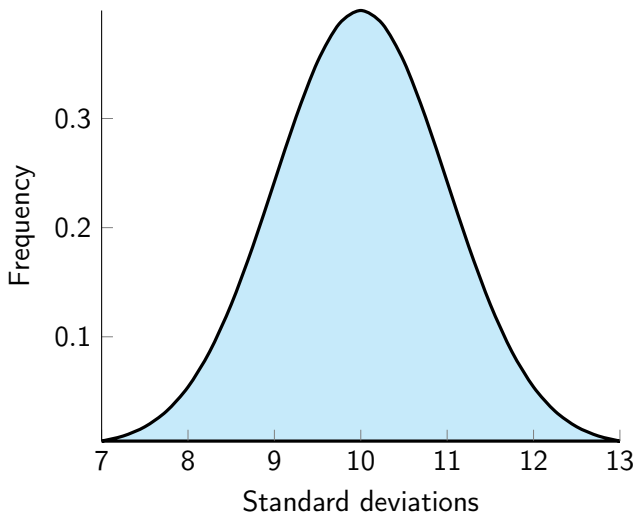
Population

μ : means

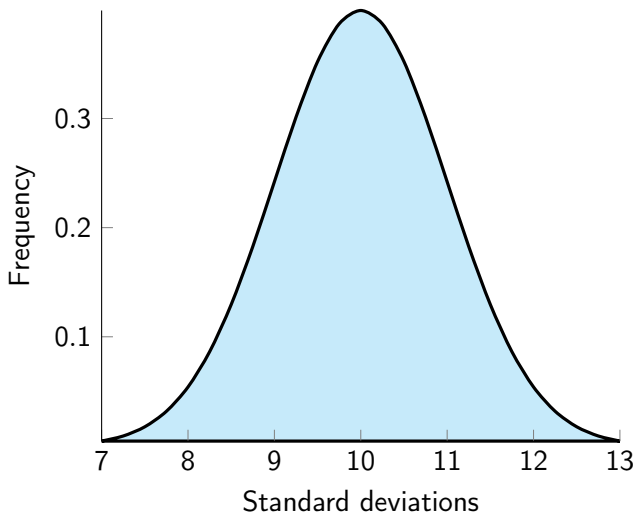
σ : standard deviation

Normal Distribution

Normal Distribution



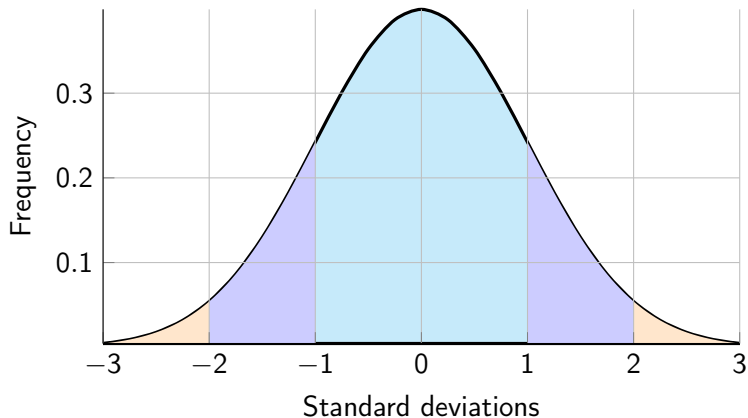
Normal Distribution



$$\bar{x} = 10$$

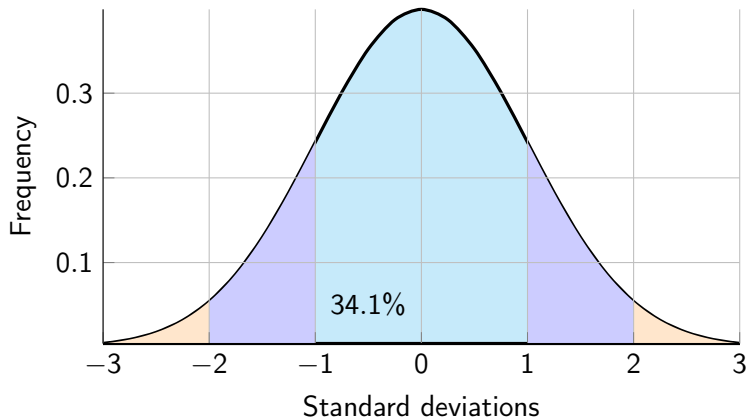
$$s = 1$$

Normal Distribution



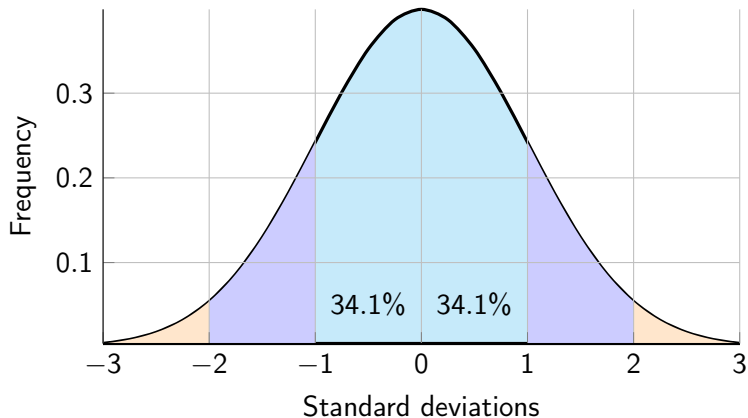
$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

Normal Distribution



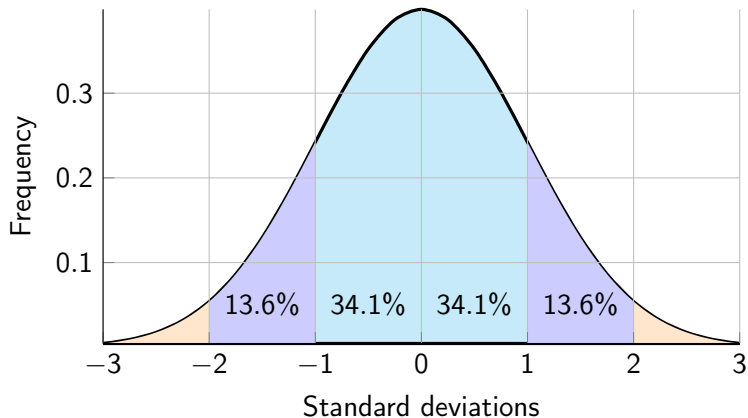
$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

Normal Distribution



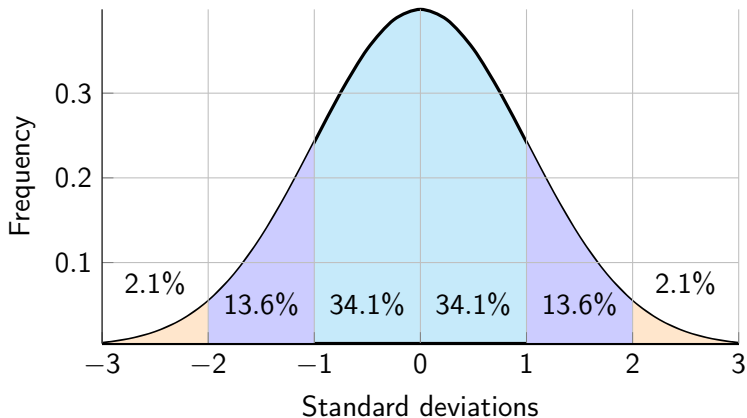
$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

Normal Distribution



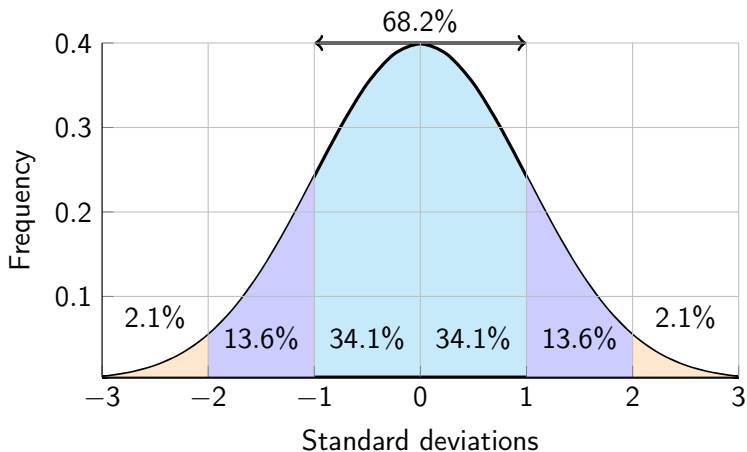
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Normal Distribution



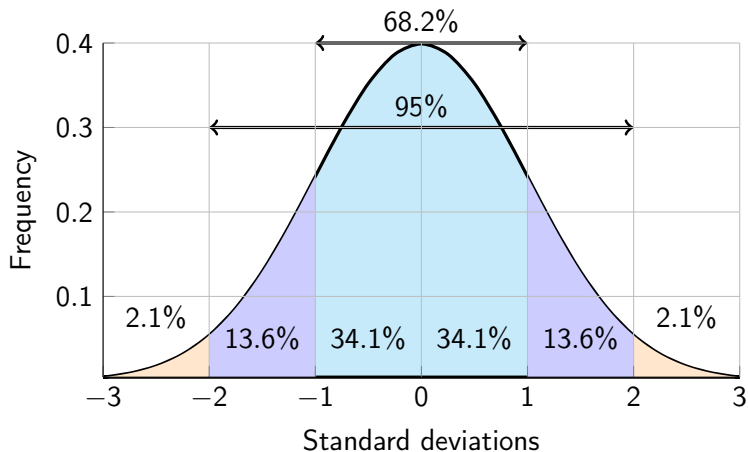
$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

Normal Distribution



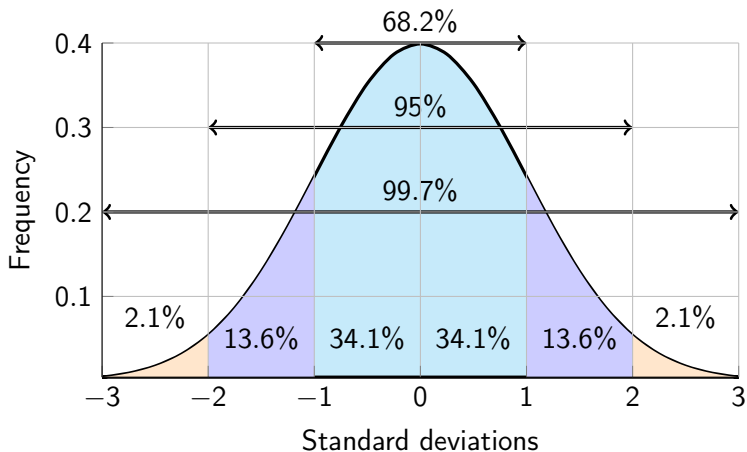
$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

Normal Distribution



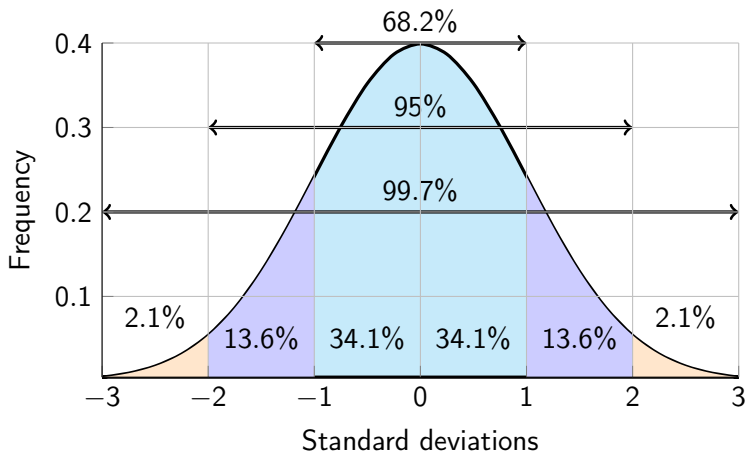
$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

Normal Distribution



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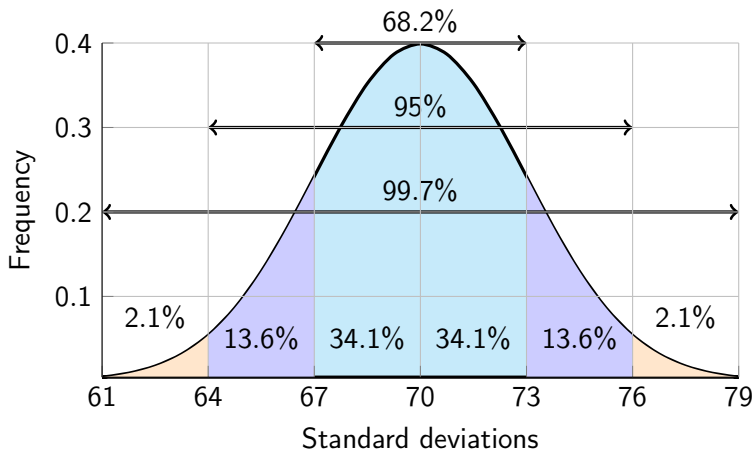
Normal Distribution



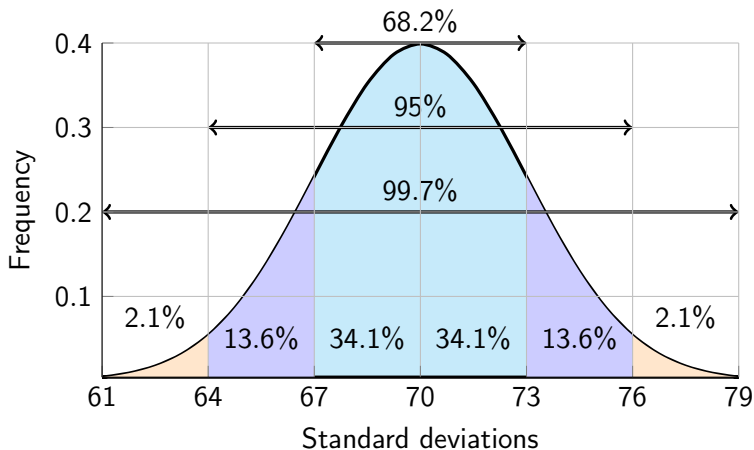
$\mu = 0$ and $\sigma = 1$ (σ for the std dev. of the *population*.)

- *Almost* all data is within 3 standard deviations of the mean (.3% on outside or 0.15% on each tail)

Normal Distribution



Normal Distribution



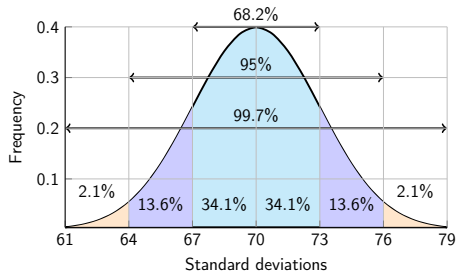
$$\mu = 70 \text{ and } \sigma = 3$$

- *Almost* all data is within 3 standard deviations of the mean

Normal Distribution

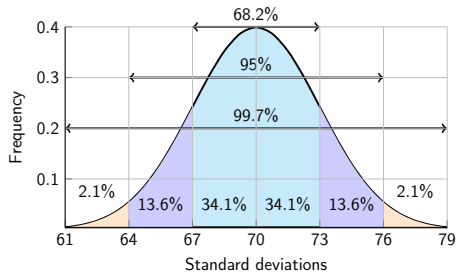
Normal Distribution

Normal Distribution



- a. 68.2%
- b. $68.2\% + 13.6\% + 2.1\% + .15\% = 84.05\%$

Normal Distribution



- a. 68.2%
- b. $68.2\% + 13.6\% + 2.1\% + .15\% = 84.05\%$
- c. $13.6\% + 2.1\% + .15\% = 15.85\%$

Normal Distribution

Normal Distribution

Example 4.4.2.

The mean height of females is 64 inches with a standard deviation of 3 inches.

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- a. What percentage of women are between 61 and 67 inches?

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The mean height of females is 64 inches with a standard deviation of 3 inches.

- a. What percentage of women are between 61 and 67 inches?
- b. What percentage of women are taller than 70 inches?
- c. Which is taller, a 70 inch woman or a 73 inch man?

Example 4.4.2.

The mean height of females is 64 inches with a standard deviation of 3 inches.

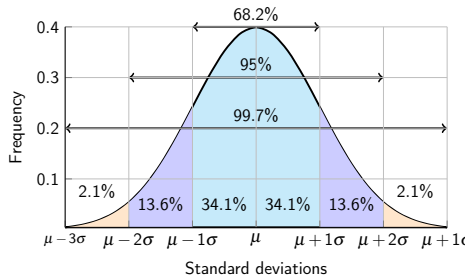
- a. What percentage of women are between 61 and 67 inches?
- b. What percentage of women are taller than 70 inches?
- c. Which is taller, a 70 inch woman or a 73 inch man?
- d. Which is more rare, a 70 inch woman or a 73 inch man?

Normal Distribution

Example 4.4.2.

The mean height of females is 64 inches with a standard deviation of 3 inches.

- What percentage of women are between 61 and 67 inches?
- What percentage of women are taller than 70 inches?
- Which is taller, a 70 inch woman or a 73 inch man?
- Which is more rare, a 70 inch woman or a 73 inch man?



- 68.2%
- $2.1\% + .15\% = 2.25\%$
- The man
- but a 70 inch woman is more rare. Relatively, she is a "taller" woman.

Normal Distribution

Normal Distribution

Example 4.4.3.

The mean height of females is 64 inches with a standard deviation of 3 inches.

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The mean height of females is 64 inches with a standard deviation of 3 inches.

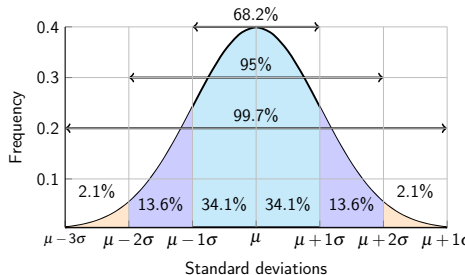
- a. Which is more rare, a 65 inch woman or a 71 inch man?

Normal Distribution

Example 4.4.3.

The mean height of females is 64 inches with a standard deviation of 3 inches.

- a. Which is more rare, a 65 inch woman or a 71 inch man?



- a. The woman